Wired network (Ethernet) troubleshooting at your desk

Article KB0010023 | Owner: Network Support | Subcategory: Ethernet / LAN | Applies to: Windows 11 laptops (Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840), macOS laptops (MacBook Pro/Air), Desktop PCs with built-in NIC, USB-C/Ethernet docking stations

Summary

This article provides step-by-step guidance for diagnosing and fixing wired Ethernet connectivity problems at an employee's desk. It covers common symptoms, initial checks, a self-help procedure users can follow, and the corresponding actions for service-desk agents. Follow the verification steps to confirm restoration before closing the ticket.

Symptoms

- No network icon or a red 'X' over the Ethernet adapter in network settings
- 'Network cable unplugged' or 'Limited or no connectivity' message
- Unable to access intranet or internet sites; ping fails
- Docking-station LAN port shows no link LED
- Link speed stuck at 100 Mbps instead of expected 1 Gbps

Common causes

- Loose or damaged Ethernet cable
- Faulty desk wall port or patch-panel mis-patch
- Defective docking-station LAN port or outdated firmware/driver
- Switch port disabled, error-disabled, or VLAN mismatch
- NIC driver issue or power-management setting

Initial Checks

Verify physical link activity by looking for LEDs on the laptop NIC, docking station, or wall jack. Ensure the Ethernet cable is firmly seated at both ends and inspect for visible damage. If a spare patch cord is available, swap it in to rule out a bad cable. Finally, disable Wi-Fi temporarily so the wired adapter is the only active network interface, making it easier to see whether the OS reports a connection.

Self-Help Troubleshooting Steps

Follow these steps if the initial checks do not restore connectivity. They are safe to perform on Acme-managed laptops and docks and require no extra administrative privileges.

1. Disconnect the Ethernet cable from both the laptop/dock and the wall jack, inspect the connectors for bent pins or debris, then reconnect firmly.
2. Check the link LEDs on the NIC/dock and wall jack; if no light appears, try a different Ethernet cable from your desk's spare kit.
3. Plug the cable into another active wall port (if available) or into a known-good port on a colleague's dock to isolate whether the fault lies with the cable, the dock/NIC, or the wall jack.
4. Update the Ethernet driver: open Device Manager -> Network adapters, right-click the Ethernet controller, select Update driver -> Search automatically. For docks, launch the vendor's docking-station utility (e.g., Dell Display Manager, Lenovo Vantage) and apply any firmware updates.
5. Reset the TCP/IP stack: open an elevated Command Prompt and run `netsh int ip reset`, `netsh winsock reset`, then reboot the computer.

Service Desk Agent Procedure

When a user reports a wired-network issue, the service-desk agent should first verify the symptom, then perform remote checks and, if needed, coordinate with the Network Support team to inspect the switch port and patch panel.

1. Open the ticket, confirm the user's location, device type, and whether Wi-Fi is disabled as per the self-help guide.
2. Ask the user to run `ipconfig /all` and note the Ethernet adapter's MAC address, link speed, and any error messages (e.g., 'Media disconnected').

3. Using the network management console, locate the switch port associated with the wall jack (via the port-to-patch-panel map) and check its status: look for link, speed/duplex, error-disabled state, or port-security violations.
4. If the port shows no link, request a physical inspection: verify patch-panel termination, swap the patch cord, or move the user to a neighboring port to test.
5. If the port is administratively shut down or error-disabled, re-enable it after confirming no loop or security violation; otherwise, replace the faulty patch cord or dock and retest.

Verification and Next Steps

After the suspected cause has been addressed, verify that the user regains network access and that the connection is stable. Document the resolution steps and any parts replaced, then close the ticket according to SLA timelines. If the problem persists, escalate as described in the Escalation section.

1. Ask the user to open a browser and navigate to an internal site (e.g., 'https://intranet.acme.example') and an external site (e.g., 'https://www.microsoft.com').
2. Run a ping test to the default gateway and to an external IP (e.g., 8.8.8.8) to confirm latency and packet loss are within normal ranges.
3. Check the link speed in the adapter properties to ensure it is negotiating at 1 Gbps full duplex.
4. If all tests pass, close the ticket with a resolution note detailing the action taken (e.g., 'Replaced faulty Ethernet cable, link restored').

When to escalate

Escalate to the Network Support team if the switch port remains down after cable and dock replacement, if the port is error-disabled due to a suspected loop or port-security violation, or if multiple users in the same area report similar issues. Include the wall-jack ID, switch port number, MAC address of the affected device, results of the ping and link-speed tests, and any relevant switch log snippets.

Keywords: Ethernet, LAN, wired network, network cable unplugged, no network, limited connectivity, docking station LAN, desk network port dead, link speed 100Mbps, switch port, patch cable, NIC driver, TCP/IP reset